

SLC35A1-Congenital Disorder of Glycosylation (SLC35A1-CDG / CDG-IIf): Comprehensive Disease Characterization

Disease: SLC35A1-Congenital Disorder of Glycosylation **MONDO ID:** MONDO:0011342 · **Category:** Mendelian (autosomal recessive) **Gene:** *SLC35A1* (Golgi CMP-sialic acid transporter; HGNC:10940, chromosome 6q15) · **OMIM phenotype:** CDG-IIf **Report date:** 2026-09-14 · Evidence base: human clinical case reports, cellular/biochemical studies, and one conditional mouse model.

Summary

SLC35A1-CDG, historically designated **CDG type IIf**, is an ultra-rare autosomal recessive congenital disorder of glycosylation caused by biallelic loss-of-function of ***SLC35A1***, the gene encoding the **sole Golgi CMP-sialic acid transporter**. This transporter imports the activated sialic acid donor (CMP-Neu5Ac) from the cytosol into the Golgi lumen, where it is the substrate for the sialyltransferases that cap N-glycans, O-glycans, and glycolipids with terminal sialic acid. When the transporter fails, the Golgi is deprived of its sialic acid donor and the cell produces **globally hyposialylated glycoproteins and glycolipids**, despite normal cytosolic sialic acid synthesis. This single upstream lesion propagates into two clinically distinct but overlapping presentations.

The disease spectrum runs from a **hematologic phenotype** — macrothrombocytopenia, bleeding tendency, neutropenia, recurrent infections, and complete absence of the sialyl-Lewis-x leukocyte ligand — to a **severe neurological phenotype** — developmental delay, hypotonia, seizures, and encephalopathy. The index patient (2005) presented with the hematologic picture; the third reported patient (2017) presented with a neurological-dominant picture and *no* hematological abnormalities, establishing genotype-dependent phenotypic heterogeneity. Mechanistically, the thrombocytopenia is explained by hepatic Ashwell-Morell receptor clearance of desialylated platelets plus impaired megakaryocytopoiesis

(demonstrated in a conditional mouse model), while the neurological phenotype is linked to deficient synthesis of the two major brain sialoglycan families — **gangliosides** and **polysialic acid on NCAM**.

Diagnosis rests on a **Type II serum transferrin glycoprofile** (isoelectric focusing, capillary zone electrophoresis, or high-resolution intact-transferrin mass spectrometry) followed by molecular confirmation of biallelic *SLC35A1* variants. **No targeted or curative therapy exists.** Unlike some other CDG subtypes treatable by monosaccharide supplementation (mannose for MPI-CDG, galactose for SLC35A2-CDG/PGM1-CDG, fucose for SLC35C1-CDG), sialic acid supplementation cannot bypass an *SLC35A1* defect because the lesion is in Golgi *delivery* of the activated donor, not in sialic acid availability. Management is supportive. Sialidase (neuraminidase) inhibition is a mechanistically plausible but unproven candidate strategy for the desialylation-driven thrombocytopenia.

1. Disease Information

Overview. SLC35A1-CDG is a Type II congenital disorder of glycosylation in which defective transport of CMP-sialic acid into the Golgi apparatus produces global hyposialylation of glycoconjugates. It is a defect of glycan *processing/elaboration* (hence "Type II"), distinct from the Type I CDGs that impair assembly of the lipid-linked oligosaccharide precursor.

Key identifiers:

Resource	Identifier
MONDO	MONDO:0011342
OMIM (phenotype)	CDG-IIf
Gene	<i>SLC35A1</i> , HGNC:10940; chromosome 6q15
Legacy name	CDG-IIf; CMP-sialic acid transporter deficiency
Classification	Congenital disorder of glycosylation, Type II

Synonyms / alternative names: CDG-IIf; CDG type IIf; CMP-sialic acid transporter deficiency; congenital disorder of glycosylation type 2F; SLC35A1-CDG.

Information source. The disease-level characterization here is derived from **aggregated disease-level resources** (OMIM, published case reports, functional studies, mouse models) rather than from individual EHR data. The clinical picture is assembled from a very small number of individually reported patients (fewer than ten worldwide since 2005), so most claims rest on single-case or small-case-series evidence supplemented by mechanistic model-organism and in-vitro work.

2. Etiology

Primary cause — genetic. SLC35A1-CDG is caused exclusively by **biallelic (compound heterozygous or homozygous) loss-of-function variants in SLC35A1**. There is no environmental or infectious etiology; the disease is a monogenic inborn error of metabolism. The original 2005 report identified a patient lacking sialyl-Lewis-x on polymorphonuclear cells who carried compound heterozygous *SLC35A1* defects: one allele with a double microdeletion producing a premature stop at codon 327, the other with a splice mutation causing a 130-bp deletion and a premature stop at codon 684. Complementation studies in Lec2 cells (which lack the CMP-sialic acid transporter) showed that neither patient allele restored sialylation, whereas wild-type transcript fully restored it — establishing loss of function as causal. *"The inactivation of one patient allele by a double microdeletion inducing a premature stop codon at position 327 and a splice mutation of the other allele inducing a 130-base pair (bp) deletion and a premature stop codon at position 684 are proposed to be the causal defects of this disease."* ([PMID: 15576474](#)).

Genetic risk factors. The only risk factor is inheritance of two pathogenic *SLC35A1* alleles. Heterozygous carriers are unaffected. As an autosomal recessive Mendelian disorder, **consanguinity** and **founder effects** increase risk in specific families/populations, although the disorder is too rare for population-specific founder alleles to have been formally established.

Environmental risk factors / protective factors / gene-environment interactions. Not applicable. No environmental risk factors, protective factors, lifestyle factors, or gene-environment interactions are known or expected for this fully penetrant monogenic disorder. No protective modifier alleles have been reported.

3. Phenotypes

SLC35A1-CDG presents along a spectrum with two overlapping symptom clusters. The index CDG-IIc patient presented with **macrothrombocytopenia, neutropenia, and complete lack of sialyl-Lewis-x** ([PMID: 27387429](#)). A subsequently reported (third) patient carried compound heterozygous missense variants p.Thr156Arg and p.Glu196Lys and presented with a **profound neurological phenotype (encephalopathy) without hematological abnormalities** ([PMID: 28856833](#)).

Phenotype	Type	HPO term (suggested)	Onset	Severity / course
Macrothrombocytopenia	Laboratory / hematologic	HP:0011897 (thrombocytopenia); HP:0040326 (giant platelets)	Congenital/neonatal	Variable; may be presenting feature
Bleeding tendency	Clinical sign	HP:0001892 (abnormal bleeding)	Congenital	Variable
Neutropenia	Laboratory	HP:0001875	Congenital/neonatal	Variable
Recurrent infections	Clinical	HP:0002719	Infancy	Related to neutropenia / absent sLe ^x
Absent sialyl-Lewis-x	Laboratory/biochemical	—	Congenital	Constant (biochemical hallmark)
Developmental delay	Clinical	HP:0001263	Infancy	Severe in neurological form
Hypotonia	Clinical sign	HP:0001252	Neonatal/infancy	Common
Seizures	Clinical sign	HP:0001250	Infancy	Present in encephalopathic form
Encephalopathy	Clinical	HP:0001298	Infancy	Severe, progressive

Phenotype characteristics. Onset is **congenital/neonatal to early infancy**. Severity is **variable and genotype-influenced**: null/truncating biallelic genotypes are associated with the hematologic-dominant picture, whereas hypomorphic missense genotypes (e.g., T156R/E196K) have been associated with the severe neurological-dominant picture. Frequency data across affected individuals cannot be quantified reliably given the very small number of reported patients; features are described qualitatively.

Quality of life impact. In the neurological form, encephalopathy, seizures, and severe developmental delay produce profound impairment of daily functioning and lifelong dependency. In the hematologic form, bleeding risk and infection susceptibility dominate. Formal QoL instruments (EQ-5D, SF-36, PROMIS) have not been applied to this ultra-rare disorder.

4. Genetic / Molecular Information

Causal gene. *SLC35A1* (solute carrier family 35 member A1; HGNC:10940), encoding the Golgi CMP-sialic acid transporter, a multipass transmembrane nucleotide-sugar transporter of the SLC35 family.

Pathogenic variants reported:

Variant	Type	Consequence	Clinical association
Double microdeletion → stop at codon 327	Frameshift/truncating	Loss of function	Index case, hematologic (PMID: 15576474)
Splice mutation → 130-bp deletion, stop at codon 684	Splice/truncating	Loss of function	Index case, hematologic (PMID: 15576474)
p.Thr156Arg (T156R)	Missense	Reduced transport	Third case, neurological (PMID: 28856833)
p.Glu196Lys (E196K)	Missense	Reduced transport; disrupts ST3Gal4 coupling	Third case, neurological (PMID: 28856833; PMID: 36257191)

Variant classification. Reported variants are pathogenic/likely pathogenic per ACMG criteria, supported by functional complementation assays. Population allele frequencies (gnomAD) for pathogenic *SLC35A1* alleles are extremely low, consistent with an ultra-rare recessive disorder.

Functional consequences. All disease variants are **loss-of-function** (either complete via truncation or partial/hypomorphic via missense reduction of transport activity). Structure-guided mutagenesis has defined three substrate pockets in *SLC35A1* — **nucleobase (E52, K55, Y214)**, **middle (Q101, N102, T260)**, and **sugar (K124, T128, S188, K272)** — with Y214 discriminating cytosine from uracil. *"The pockets comprise (1) nucleobase (residues E52, K55,*

and Y214 of SLC35A1..." (PMID: 34384782). Beyond transport, SLC35A1 physically associates with the α 2,3-sialyltransferase **ST3Gal4**; the CDG-causing **E196K mutation (but not T156R) disrupts this interaction** and E196K is less efficient at restoring N-glycan sialylation in SLC35A1-knockout cells: *"This phenomenon is compromised by the E196K (but not T156R) mutation in the SLC35A1 gene. We also demonstrated that the E196K mutant is less efficient in restoring N-glycan sialylation upon expression in the SLC35A1 knockout cells."* (PMID: 36257191). This shows that some missense variants impair sialylation both by reducing transport and by decoupling the transporter from the sialyltransferase machinery.

Modifier genes, epigenetics, chromosomal abnormalities. No modifier genes, epigenetic mechanisms, or chromosomal-scale abnormalities have been established for SLC35A1-CDG. Origin is germline; somatic contribution is not applicable.

5. Environmental Information

Not applicable. SLC35A1-CDG is a purely monogenic disorder. No environmental factors, toxins, lifestyle factors, or infectious agents cause or trigger the disease. Recurrent infections in affected individuals are a *consequence* of neutropenia and absent leukocyte sialyl-Lewis-x adhesion ligands, not an etiologic agent.

6. Mechanism / Pathophysiology

Ordered causal chain

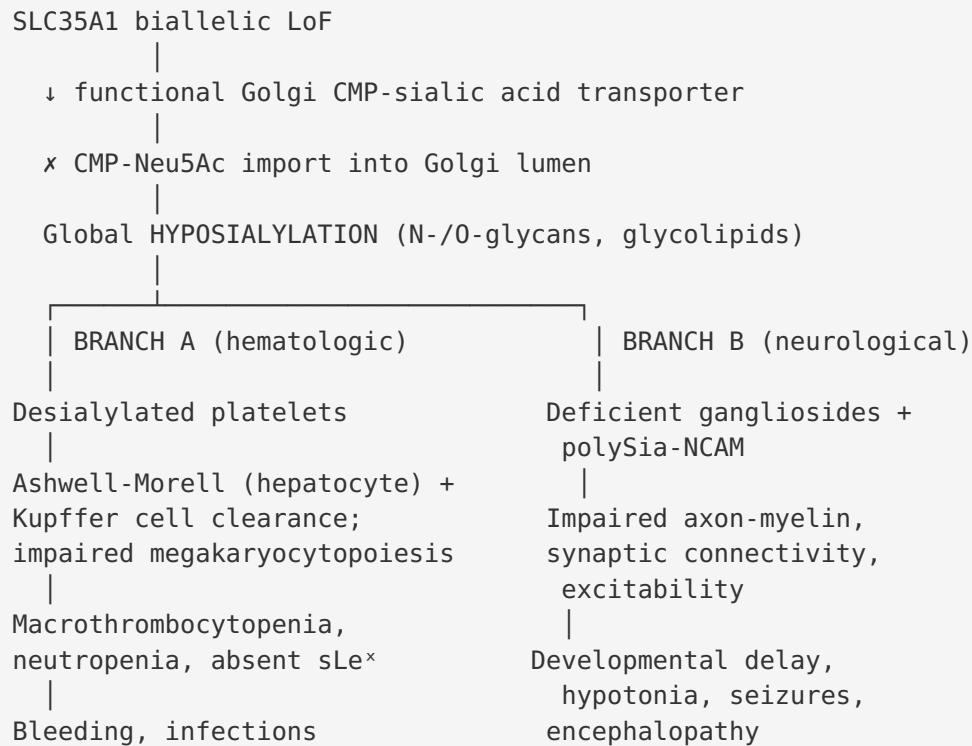
1. **Biallelic loss-of-function variants in SLC35A1** → **loss/reduction of functional Golgi CMP–sialic acid transporter** protein (demonstrated by complementation failure in Lec2 cells; PMID: 15576474).
2. Loss of transporter → **failure to import CMP-Neu5Ac (the activated sialic acid donor) from cytosol into the Golgi lumen** (direct: reduced Golgi CMP-sialic acid transport rate measured in patient fibroblasts; PMID: 28856833).
3. Golgi donor depletion → **sialyltransferases (e.g., ST3Gal4) cannot cap glycans**, leading to **global hyposialylation of N-glycans, O-glycans, and glycolipids** (direct: decreased sialylated N-/O-glycans in patient serum and fibroblasts; PMID: 28856833). Some missense

variants (E196K) additionally **decouple SLC35A1 from ST3Gal4**, worsening the sialylation defect ([PMID: 36257191](#)).

4. Hyposialylation then **branches** into tissue-specific consequences:

Branch A — Hematologic: - 4A-i. Loss of terminal sialic acid on platelet surface glycoproteins **exposes subterminal galactose** → recognition by **hepatocyte Ashwell-Morell receptors** → **Fc-independent hepatic platelet clearance** (inferred for SLC35A1-CDG from the general desialylation-clearance mechanism; demonstrated directly in ITP models; [PMID: 26185093](#)). - 4A-ii. In parallel, hyposialylation **impairs megakaryocytopoiesis** — reduced bone marrow megakaryocyte numbers and impaired maturation — with excess desialylated platelets cleared by **hepatic Kupffer cells** (direct, mouse model; [PMID: 32303557](#)). - 4A-iii. → **Macrothrombocytopenia, bleeding, neutropenia, absent leukocyte sialyl-Lewis-x** → **recurrent infections**.

Branch B — Neurological: - 4B-i. Loss of Golgi sialic acid delivery **impairs synthesis of the two major brain sialoglycan families** — **gangliosides and polysialic acid on NCAM** (inferred from the essential requirement of these glycans for sialic acid; [PMID: 24692354](#)). - 4B-ii. Ganglioside deficiency **disrupts axon-myelin interactions, axon stability/regeneration, and neuronal excitability**; polySia-NCAM deficiency **disrupts neurite outgrowth, synaptic connectivity, and memory formation** ([PMID: 24692354](#); [PMID: 22585926](#)). - 4B-iii. → **Developmental delay, hypotonia, seizures, encephalopathy**.



Detail by category

- **Molecular pathways / biochemical abnormality.** The core defect is in the **sialylation branch of the glycosylation pathway** (KEGG/Reactome: "Sialic acid metabolism"; Reactome "Transport of nucleotide sugars"). The specific biochemical lesion is **failure of CMP-sialic acid (CMP-Neu5Ac) antiport into the Golgi**, upstream of all sialyltransferase reactions.
- **Protein dysfunction.** Loss of function of a multipass Golgi membrane transporter, via truncation or hypomorphic missense changes mapping to defined substrate pockets; some missense variants also disrupt a protein-protein interaction with ST3Gal4.
- **Cellular processes.** Impaired megakaryocyte maturation (hematopoiesis); increased platelet clearance (phagocytosis/endocytic clearance); impaired neuronal differentiation/connectivity.
- **Metabolic changes.** Altered glycoconjugate metabolism — deficiency of sialylated glycolipids (gangliosides) and glycoproteins; cytosolic sialic acid synthesis remains intact.
- **Immune involvement.** Secondary immunodeficiency: neutropenia plus loss of sialyl-Lewis-x (the selectin ligand mediating leukocyte rolling/adhesion) impairs leukocyte trafficking → recurrent infections.

- **Molecular profiling.** Patient serum and fibroblast glycomics show markedly decreased N- and O-glycans terminating in sialic acid ([PMID: 28856833](#)); intact-transferrin high-resolution mass spectrometry detects a Type II pattern ([PMID: 26307094](#)).

Suggested GO / CL terms. GO:0015739 (sialic acid transport), GO:0008373 (sialyltransferase activity), GO:0006486 (protein glycosylation), GO:0005794 (Golgi apparatus), GO:0000139 (Golgi membrane). Cell types (CL): CL:0000556 (megakaryocyte), CL:0000233 (platelet), CL:0000775 (neutrophil), CL:0000540 (neuron).

7. Anatomical Structures Affected

Organ / system level. - **Hematopoietic / blood system** (UBERON:0002390 hematopoietic system; UBERON:0000178 blood): platelets, megakaryocytes, neutrophils. - **Liver** (UBERON:0002107): site of desialylated-platelet clearance via Ashwell-Morell receptors (hepatocytes) and Kupffer cells. - **Central nervous system / brain** (UBERON:0000955): primary target in the neurological form. - **Bone marrow** (UBERON:0002371): impaired megakaryocytopoiesis.

Tissue and cell level. Megakaryocytes (CL:0000556) and platelets (CL:0000233); neutrophils/polymorphonuclear cells (CL:0000775); neurons and glia in the CNS; hepatocytes (CL:0000182) and Kupffer cells (CL:0000091) as clearance sites.

Subcellular level. The primary lesion is at the **Golgi apparatus membrane** (GO:0000139 Golgi membrane; GO:0005794 Golgi apparatus). Downstream, glycolipid/ganglioside deficits affect **plasma membrane** composition (GO:0005886).

Localization / lateralization. Systemic and bilateral; no lateralization. The disorder is multisystem, reflecting the ubiquitous requirement for sialylation.

8. Temporal Development

- **Onset:** Congenital; clinical manifestations appear in the **neonatal period to early infancy**. Onset pattern is chronic/insidious for developmental features and can be acute for bleeding or infection episodes.

- **Progression:** In the neurological form, encephalopathy and developmental impairment are severe and effectively lifelong; disease course is progressive to stable-severe. In the hematologic form, thrombocytopenia and infection susceptibility persist chronically.
 - **Duration:** Chronic, lifelong.
 - **Critical periods:** Early infancy is the window in which neurodevelopmental sialoglycan requirements (ganglioside and polySia-NCAM-dependent synaptogenesis) are greatest, making this the theoretical window of vulnerability and, hypothetically, of intervention.
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9. Inheritance and Population

- **Inheritance:** Autosomal recessive (biallelic *SLC35A1* loss-of-function; [PMID: 15576474](#)).
 - **Penetrance / expressivity:** Effectively complete penetrance in biallelic carriers; **variable expressivity** across the hematologic–neurological spectrum, influenced by genotype (null vs hypomorphic missense).
 - **Epidemiology: Ultra-rare.** Only a handful of patients reported worldwide since 2005; the 2017 encephalopathy report described its subject as *"the third patient with CMP-sialic acid transporter deficiency"* ([PMID: 28856833](#)). Precise prevalence/incidence cannot be estimated. For context, aggregate CDG screening in Tunisia over 15 years estimated total CDGS incidence at ~1:23,720 live births (4.21/100,000), of which *SLC35A1*-CDG would be a minute fraction ([PMID: 38262859](#)).
 - **Carrier frequency / founder effects:** Not established; extremely low based on gnomAD rarity of pathogenic alleles. Consanguinity would increase homozygous risk.
 - **Demographics:** No sex predilection (autosomal); no established ethnic or geographic clustering given the small number of cases.
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10. Diagnostics

First-line biochemical screen. Plasma **transferrin glycoform analysis** — isoelectric focusing (IEF), capillary zone electrophoresis (CZE), or high-resolution mass spectrometry. *SLC35A1*-CDG produces a **Type II (CDG-II) transferrin pattern**, reflecting defective glycan sialylation/processing rather than whole-glycan loss. **High-resolution intact-transferrin mass spectrometry** can directly detect and subtype the *SLC35A1*-CDG defect among CDG-II

disorders; SLC35A1-CDG is explicitly listed among "*Known CDG-II defects (phosphoglucomutase 1 [PGM1-CDG], ... [MGAT2-CDG], ... [B4GALT1-CDG], CMP-sialic acid transporter [SLC35A1-CDG]...*" (PMID: 26307094).

Confirmatory glycomics. Serum/fibroblast N- and O-glycan profiling shows markedly decreased sialylated species; assay of Golgi CMP-sialic acid transport rate in patient fibroblasts is reduced (PMID: 28856833).

Specialized markers. Absent/decreased **sialyl-Lewis-x** on polymorphonuclear cells is a distinctive biochemical hallmark in the hematologic form (PMID: 27387429).

Hematology. CBC with platelet indices (macrothrombocytopenia, giant platelets), neutrophil count (neutropenia), and peripheral smear.

Genetic testing. Molecular confirmation of **biallelic SLC35A1 variants** is definitive. Recommended approach: exome or genome sequencing, or a targeted CDG/glycosylation gene panel that includes *SLC35A1*; complementation assays (as in Lec2 cells) can functionally confirm novel variants (PMID: 15576474). Chromosomal microarray, karyotyping, FISH, mtDNA, and repeat-expansion testing are **not applicable**.

Clinical criteria / differential diagnosis. No disease-specific consensus criteria exist; diagnosis follows the general CDG workup (abnormal transferrin glycoform → glycan analysis → molecular confirmation). Differential diagnosis includes other CDG-II subtypes (PGM1-CDG, MGAT2-CDG, B4GALT1-CDG, SLC35A2-CDG), inherited thrombocytopenias/macrothrombocytopenias, and other genetic encephalopathies.

Screening. Not part of routine newborn screening. Carrier and cascade testing are appropriate in families with a known pathogenic genotype.

11. Outcome / Prognosis

- **Survival/mortality:** Formal survival statistics are unavailable given ultra-rarity. The severe neurological form carries a guarded prognosis with substantial morbidity; the hematologic form's prognosis is dominated by bleeding and infection risk.
- **Morbidity/disability:** In the neurological form, profound and lifelong developmental disability, seizures, and encephalopathy. In the hematologic form, bleeding tendency and infection susceptibility.

- **Complications:** Hemorrhage (thrombocytopenia), recurrent/severe infections (neutropenia + adhesion defect), seizures, and developmental sequelae.
- **Recovery potential:** Limited; the underlying transport defect is not correctable, so outcomes reflect symptomatic burden.
- **Prognostic factors:** Genotype (null vs hypomorphic), predominant organ system, seizure control, and infection/bleeding management. No validated prognostic biomarkers exist.

12. Treatment

No targeted or curative therapy exists. Management is **supportive and symptomatic:** - **Hematologic support:** transfusion and bleeding precautions for thrombocytopenia; infection prophylaxis/treatment and management of neutropenia. - **Neurological support:** anti-seizure medication; developmental, physical, occupational, and speech therapy; supportive/palliative care. - **Nutrition:** general CDG nutritional support.

Why monosaccharide supplementation does not work here. Reviews of CDG therapeutics emphasize that among 160+ CDG subtypes few have specific therapy, and available treatments are largely **dietary monosaccharide/precursor supplementation** — e.g., mannose (MPI-CDG/PMM2-CDG trials), galactose (SLC35A2-CDG, PGM1-CDG), fucose (SLC35C1-CDG). *"Patients present with a wide range of symptoms and therapies are only available for very few subtypes. Specific nutritional treatment options for certain CDG types include oral supplementation of monosaccharide sugars, manganese, uridine, or pyridoxine."* (PMID: 35562242); *"Although the number of identified CDG is growing rapidly, there are few therapeutic options. Most treatments involve dietary supplementation with monosaccharides or other precursors."* (PMID: 34788024). For SLC35A1-CDG, **sialic acid (Neu5Ac) supplementation cannot readily bypass the defect** because the lesion is in Golgi *delivery* of the activated CMP-sialic acid donor, not in cytosolic sialic acid availability. No supplementation therapy is established for this subtype.

Candidate/experimental strategy. Because the thrombocytopenia is driven by clearance of desialylated platelets, **sialidase (neuraminidase) inhibition** is a mechanistically rational candidate: *"sialidase inhibitors ameliorate anti-GPa-mediated thrombocytopenia in mice."* (PMID: 26185093). This has **not** been tested in SLC35A1-CDG and is hypothetical. Gene therapy and other advanced therapeutics have not been developed. **Suggested NCIT terms:** platelet transfusion, supportive care, anticonvulsant therapy.

13. Prevention

- **Primary prevention:** Not applicable (monogenic). Prevention is limited to **genetic counseling and reproductive options** — carrier testing, prenatal diagnosis, and preimplantation genetic testing in families with a known genotype.
 - **Secondary/tertiary prevention:** Early diagnosis via transferrin glycoprofiling and molecular testing enables anticipatory management of bleeding, infection, and developmental support (tertiary prevention of complications).
 - **Counseling:** Autosomal recessive recurrence risk is 25% for future pregnancies of carrier couples; cascade testing of relatives is appropriate.
 - **Immunization / public health / environmental interventions:** Not applicable to the disease etiology, though standard immunizations and infection precautions are advisable given the immune compromise.
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14. Other Species / Natural Disease

- **Taxonomy / orthologues:** *SLC35A1* is conserved across mammals. The functional mouse orthologue is *Slc35a1* (*Mus musculus*, NCBI Taxon:10090). The classic in-vitro model is the CHO **Lec2** mutant, which lacks the CMP-sialic acid transporter and is used for complementation assays (PMID: [15576474](#)).
 - **Natural disease in animals:** No spontaneously occurring SLC35A1 disease has been characterized in companion animals or wildlife (no OMIA entry noted here).
 - **Comparative biology:** The transport-and-sialylation mechanism is evolutionarily conserved; the mouse conditional knockout reproduces the hematologic mechanism (see Section 15), supporting cross-species conservation of the disease mechanism.
 - **Zoonotic potential:** Not applicable.
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15. Model Organisms

Model	Type	Key features	Relevance
Platelet/megakaryocyte-specific <i>Slc35a1</i> conditional knockout mouse (Plt <i>Slc35a1</i>^{-/-})	Mammalian, conditional KO	Thrombocytopenia; reduced/immature bone marrow megakaryocytes; increased hepatic Kupffer-cell clearance of desialylated platelets	Recapitulates the hematologic mechanism (PMID: 32303557)
CHO Lec2 mutant cells	In vitro cell line	Deficient CMP-sialic acid transporter	Complementation/functional assay platform (PMID: 15576474)
SLC35A1-knockout cell lines	In vitro	Loss of N-glycan sialylation; used to test rescue by variant constructs	Variant functional characterization (PMID: 36257191)
Patient-derived fibroblasts	In vitro (human)	Decreased sialylated N-/O-glycans; reduced Golgi CMP-sialic acid transport rate	Biochemical confirmation (PMID: 28856833)

Phenotype recapitulation & limitations. The conditional mouse faithfully models the **hematologic** arm (thrombocytopenia via a dual defect: impaired megakaryocytopoiesis plus hepatic clearance) but, being platelet/megakaryocyte-restricted, does **not** model the neurological arm. *"The number of bone marrow megakaryocytes in Plt *Slc35a1*^{-/-} mice was reduced, and megakaryocyte maturation was also impaired. In addition, an increased number of desialylated platelets was cleared by Kupffer cells in the liver of Plt *Slc35a1*^{-/-} mice."* (PMID: [32303557](#)). Cell-based systems capture the biochemical sialylation defect and enable variant functional testing but cannot reproduce the multisystem clinical phenotype. A neural or whole-body model that recapitulates the encephalopathy has not been reported — a notable gap.

Key Findings (Evidence Detail)

Finding 1 — SLC35A1-CDG is caused by biallelic loss of the Golgi CMP-sialic acid transporter

The 2005 index report identified a patient lacking sialyl-Lewis-x on PMN cells with compound heterozygous *SLC35A1* defects (double microdeletion → stop at codon 327; splice mutation → 130-bp deletion, stop at codon 684). Complementation in Lec2 cells showed neither patient allele restored sialylation while wild-type did. *"We conclude that this defect is a new type of congenital disorder of glycosylation (CDG) of type II affecting the transport of CMP-sialic acid into the Golgi apparatus."* (PMID: 15576474)

Finding 2 — Two overlapping presentations: hematologic vs neurological

Hematologic-dominant (macrothrombocytopenia, neutropenia, absent sLe^x) in the index case (PMID: 27387429); neurological-dominant encephalopathy with missense genotype (T156R/E196K) and no hematologic abnormality in the third patient: *"Here we report the identification of the third patient with CMP-sialic acid transporter deficiency, who presented with severe neurological phenotype, but without hematological abnormalities."* Patient fibroblasts/serum showed *"a considerable decrease in the amount of N- and O-glycans terminating in sialic acid"* (PMID: 28856833).

Finding 3 — Mouse Slc35a1 deficiency: thrombocytopenia via impaired megakaryocytopoiesis + hepatic clearance

A conditional platelet/megakaryocyte-specific knockout produced thrombocytopenia via reduced megakaryocyte numbers, impaired maturation, and increased Kupffer-cell clearance of desialylated platelets (PMID: 32303557).

Finding 4 — Substrate pockets and a sialyltransferase interaction disrupted by CDG mutations

Three substrate pockets defined (nucleobase E52/K55/Y214; middle Q101/N102/T260; sugar K124/T128/S188/K272) (PMID: 34384782). E196K (not T156R) disrupts SLC35A1–ST3Gal4 coupling and reduces sialylation rescue in knockout cells (PMID: 36257191).

Finding 5 — Ultra-rare; diagnosis via Type II transferrin glycoprofile + molecular confirmation

Only the third patient by 2017 ([PMID: 28856833](#)); classified among CDG-II defects detectable by intact-transferrin mass spectrometry alongside PGM1-, MGAT2-, and B4GALT1-CDG ([PMID: 26307094](#)).

Finding 6 — No targeted therapy; supportive management within the CDG landscape

Therapy is available for very few CDG subtypes and is largely dietary monosaccharide/precursor supplementation ([PMID: 35562242](#); [PMID: 34788024](#)). Sialic acid supplementation cannot bypass the Golgi-delivery lesion.

Finding 7 — Neurological phenotype linked to deficient brain sialoglycans

"In the brain, two families of sialoglycans are of particular interest: gangliosides and polysialic acid." and *"Mouse genetic studies and human disorders of ganglioside metabolism implicate gangliosides in axon-myelin interactions, axon stability, axon regeneration, and the modulation of nerve cell excitability."* ([PMID: 24692354](#)); *"Polysialylated NCAM and neural gangliosides both play critical roles in mediating cell-to-cell interactions important for neuronal outgrowth, synaptic connectivity, and memory formation."* ([PMID: 22585926](#)).

Finding 8 — Ashwell-Morell receptor clearance underlies thrombocytopenia; sialidase inhibition is a candidate therapy

"This leads to platelet clearance in the liver via hepatocyte Ashwell-Morell receptors, which is fundamentally different from the classical Fc-FcγR-dependent macrophage phagocytosis." and *"sialidase inhibitors ameliorate anti-GPa-mediated thrombocytopenia in mice."* ([PMID: 26185093](#)).

Mechanistic Model / Interpretation

SLC35A1-CDG is a paradigm of **"one lesion, two phenotypes."** A single upstream defect — failure to deliver CMP-sialic acid into the Golgi — produces global hyposialylation, which then manifests differently depending on which sialylated glycoconjugates a given tissue most

depends on. In the **blood**, the critical cargo is sialic acid on platelet and leukocyte surface glycoproteins: its loss triggers hepatic clearance of platelets (Ashwell-Morell + Kupffer) and abolishes the selectin ligand sialyl-Lewis-x, producing thrombocytopenia, bleeding, and infection susceptibility. In the **brain**, the critical cargo is sialic acid on gangliosides and polysialic acid–NCAM: its loss impairs axon-myelin interaction, synaptic connectivity, and excitability, producing encephalopathy, seizures, and developmental delay.

The **genotype-phenotype correlation** appears to hinge on residual transporter activity: truncating null alleles (index case) associate with the hematologic picture, whereas hypomorphic missense alleles (T156R/E196K) associate with the neurological picture — possibly because partial activity, tissue-specific demand, or selective decoupling from particular sialyltransferases (E196K–ST3Gal4) shapes which organ crosses its functional threshold. This remains an inference from very few patients and should be treated cautiously.

Evidence Base

PMID	Contribution	Evidence type
15576474	Defines CDG-II _f ; biallelic LoF; Lec2 complementation	Human + in vitro
28856833	Third patient; neurological phenotype; missense genotype; reduced transport	Human + in vitro
27387429	Hematologic presentation; absent sLe ^x	Human/review
32303557	Mouse KO: megakaryocytopoiesis + hepatic clearance	Model organism
34384782	Substrate-pocket mapping	In vitro/structural
36257191	SLC35A1–ST3Gal4 interaction; E196K decoupling	In vitro
26307094	Diagnostic transferrin MS; CDG-II classification	Human/methods
24692354	Brain sialoglycans (gangliosides, polySia)	Review
22585926	Sialic acid in neurodevelopment/cognition	Review
26185093	Ashwell-Morell clearance; sialidase inhibitors	Model organism/mechanism
35562242 , 34788024	CDG therapeutics landscape	Review
38262859	CDG epidemiology context (Tunisia)	Human/epidemiology

Limitations and Knowledge Gaps

1. **Extreme rarity.** Fewer than ~10 reported patients; phenotype frequencies, penetrance details, natural history, and survival cannot be quantified.
2. **Genotype-phenotype correlation is provisional,** based on individual cases; the mechanistic explanation for why some genotypes are hematologic and others neurological is inferred, not proven.

3. **No neural model.** The mouse model is platelet/megakaryocyte-restricted and does not capture the encephalopathy; the CNS mechanism (ganglioside/polySia deficiency) is inferred from sialoglycan biology rather than demonstrated in an SLC35A1 model.
 4. **The Ashwell-Morell mechanism** in SLC35A1-CDG is extrapolated from ITP/desialylation models; direct demonstration in patient platelets is lacking.
 5. **No therapy.** No trials, no established supplementation strategy; sialidase inhibition is untested in this disorder.
 6. **Ontology term suggestions** here (HPO/GO/CL/UBERON/NCIT) are proposed based on clinical/mechanistic mapping and should be curator-verified.
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Proposed Follow-up Experiments / Actions

1. **Build a neural SLC35A1 model** (neuron-specific conditional KO mouse or patient iPSC-derived neurons/organoids) to directly test whether ganglioside and polySia-NCAM deficiency drives the encephalopathy, and to define the critical developmental window.
 2. **Systematic genotype-phenotype curation** across all reported patients, correlating residual transport activity of each variant (quantitative Lec2/KO-cell rescue assays) with predominant organ involvement.
 3. **Test sialidase inhibition** (e.g., oseltamivir-class neuraminidase inhibitors) in the Plt Slc35a1^{-/-} mouse to determine whether blocking desialylation-driven clearance corrects thrombocytopenia — a translatable candidate for the hematologic phenotype.
 4. **Evaluate whether any sialic acid-precursor or metabolic-bypass strategy** (e.g., agents raising cytosolic CMP-Neu5Ac or leveraging alternative transporters) can partially restore Golgi sialylation in patient fibroblasts.
 5. **Characterize the SLC35A1-sialyltransferase interactome** beyond ST3Gal4 to understand variant-specific decoupling and its contribution to tissue-selective phenotypes.
 6. **Establish a clinical registry / minimal biomarker panel** (transferrin glycoprofile, serum sialoglycan/ganglioside markers such as GM3, platelet desialylation markers) to enable natural-history study and future trial readiness.
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Report compiled from 5 iterations of autonomous investigation; 8 confirmed findings; 26 papers reviewed. Evidence types are distinguished throughout as human clinical, model organism, in vitro, or review.

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